PROJECT REPORT
DEEP LEARNING – DRIVEN SOIL HEALTH ASSESMENT USING CAMERA – BASED IMAGE PROCESSING

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1. ABSTRACT

Soil health assessment is essential for sustainable agriculture and improved crop productivity. Traditional soil testing methods rely on laboratory-based chemical analysis, which is time-consuming, costly, and often inaccessible to small-scale farmers. Existing automated solutions frequently require specialized sensors or hardware, limiting their practical deployment.

This project proposes a deep learning–based, camera-driven soil health assessment system that utilizes standard RGB images captured using smartphones. Convolutional Neural Networks (CNNs) are employed to extract visual features from soil images and estimate multiple soil health parameters in a non-destructive and low-cost manner. The system is designed to be robust against variations in lighting, camera quality, and environmental conditions through effective image preprocessing techniques.

To enhance user trust and transparency, explainable AI mechanisms are integrated to visualize model decision regions. The proposed solution follows a smartphone-first, edge-deployable design with offline capability, making it suitable for real-world agricultural environments. Overall, the system provides a scalable, affordable, and explainable approach to soil health assessment with strong potential for practical adoption and commercialization.

2. INTRODUCTION

Soil health plays a vital role in determining agricultural productivity, sustainability, and environmental balance. Accurate assessment of soil properties such as moisture, texture, organic matter, nutrient status, and pH is essential for informed decision-making in farming. Conventional soil testing methods rely on laboratory-based chemical analysis, which is often time-consuming, costly, and inaccessible to small and marginal farmers.

With the rapid penetration of smartphones and advances in artificial intelligence, camera-based image analysis has emerged as a promising alternative for soil assessment. Deep learning techniques, particularly Convolutional Neural Networks (CNNs), enable automatic extraction of complex visual patterns from soil images, making it possible to estimate soil health indicators without destructive testing or specialized equipment.

This project proposes **a product-oriented, deep learning–driven soil health assessment system** that leverages standard RGB camera images to deliver instant, non-destructive, and low-cost soil health insights suitable for real-world agricultural deployment.

3. PROBLEM STATEMENT

Traditional soil testing methods suffer from several limitations:

- High cost and long turnaround time due to laboratory dependency
- Destructive sampling of soil
- Limited accessibility for rural and small-scale farmers
- Inability to provide instant, field-level feedback

Although sensor-based and spectroscopy-based systems automate soil analysis, they introduce additional hardware costs and infrastructure complexity. Existing image-based solutions often focus on

isolated soil parameters, lack robustness to real-world conditions, and operate as black-box models that reduce user trust.

Hence, the problem addressed by this project is:

To design a low-cost, camera-based, deep learning system capable of providing holistic, explainable, and deployment-ready soil health assessment under real-world conditions using commonly available devices such as smartphones.

4. OBJECTIVES

- To develop an end-to-end deep learning pipeline for soil health assessment using RGB images
- To estimate multiple soil parameters from a single soil image
- To ensure robustness against lighting variations, camera differences, and environmental noise
- To integrate explainable AI mechanisms for transparent predictions
- To design the system as a scalable, smartphone-deployable product

5. TECHNICAL DOMAIN DESCRIPTION

5.1 Primary Domain

- Artificial Intelligence (Deep Learning)

5.2 Supporting Domains

- Computer Vision and Image Processing
- Agricultural Informatics
- Edge AI and Mobile Computing
- Explainable Artificial Intelligence (XAI)

5.3 Technical Overview

The proposed system captures soil surface images using a standard RGB camera. These images undergo preprocessing to normalize lighting and colour variations. A deep learning model then performs multi-task inference to estimate soil health parameters. The final output is presented through a user-friendly interface optimized for agricultural users.

6. NOVELTY OF THE PROPOSED WORK

The novelty of the proposed system lies in extending a conventional soil classification framework into a comprehensive soil health assessment pipeline without altering the core trained model. Instead of redesigning or retraining a new deep network, the work introduces architectural and methodological enhancements that improve practical applicability, interpretability, and functionality.

Unlike existing soil analysis systems that rely on specialized sensors, spectral imaging, or laboratory measurements, the proposed approach adopts a **zero-hardware, RGB-only strategy**, enabling soil health estimation using standard camera images. This significantly reduces deployment cost and improves accessibility for real-world agricultural scenarios.

The key novel contributions of this work are summarized as follows:

6.1 Pipeline Extension Without Model Modification

Most prior studies design separate deep learning models for each soil parameter (moisture, salinity, nutrients, etc.), leading to increased computational and data requirements. In contrast, the proposed system preserves a fixed EfficientNetV2-based classifier and introduces a **post-classification branching pipeline** that reuses learned visual representations for additional soil health inference.

This avoids expensive retraining and demonstrates efficient reuse of deep features.

6.2 Hybrid Deep Learning + Rule-Based Inference

Existing approaches predominantly depend on fully supervised learning, which is impractical when labelled datasets are scarce. The proposed system introduces a **hybrid inference mechanism**, where CNN-extracted features are combined with domain knowledge-driven rules to estimate soil properties.

This allows meaningful predictions even under limited data conditions — a common challenge in soil science applications.

6.3 Visual Proxy-Based Soil Health Estimation

Several soil properties (pH, organic matter, salinity) are traditionally measured via chemical analysis. The novelty here is the formulation of **visual proxy indicators** derived from colour and texture cues in RGB images, enabling approximate but actionable assessments.

This bridges the gap between laboratory-centric methods and field-level usability.

6.4 Explainability Integration for Agricultural AI

While many soil prediction models function as black boxes, the proposed system embeds **Grad-CAM-based visual explanations** directly into the prediction workflow. This enhances transparency, user trust, and diagnostic interpretability — aspects rarely emphasized in prior soil imaging studies.

6.5 Practical System Design for Low-Resource Environments

Rather than focusing purely on model accuracy, the proposed work emphasizes **deploy ability, usability, and hardware independence**, making it suitable for smartphone-based agricultural assistance systems.

Summary of Novelty

In essence, the novelty of the project is not rooted in creating a new deep learning architecture, but in:

- Re-purposing a fixed CNN for multi-parameter soil health assessment
- Introducing a hybrid deep learning + rule-based reasoning framework
- Eliminating dependence on costly sensing hardware
- Enhancing interpretability through explainable AI

This results in a scalable, low-cost, and practically viable soil health estimation system.

7. LITERATURE REVIEW

Extensive literature analysis reveals that early approaches relied on classical machine learning techniques using handcrafted features, which limited scalability and robustness. Recent CNN-based approaches improved accuracy but are often task-specific, hardware-dependent, or lack interpretability. The review highlights the following research gaps:

- Lack of holistic soil health assessment
- Dependence on expensive sensors or multi-spectral cameras
- Poor generalization across regions and lighting conditions
- Limited focus on explainability and product deployment

The proposed project addresses these gaps by adopting a unified, camera-only, explainable deep learning framework suitable for real-world use.

Table 7.1: Table Showing detailed Literature Survey of the existing research work

S. No.	Paper Title	Publisher	Methodologies	Advantages	Gaps Identified	Gap Resolution

1	Deep Learning-Based Prediction of Soil Organic Matter Using Digital Images, 2023	IEEE Xplore	CNN on RGB images	Non-destructive; low cost compared to chemical analysis.	Sensitive to ambient light variations; limited to surface SOM.	Use **Gamma Correction** & **Histogram Equalization** for lighting robustness.
2	Convolutional neural networks for predicting soil properties from RGB images, 2021	Elsevier	Multi-channel CNN	High accuracy for texture (sand/clay) prediction.	Requires large, diverse datasets for generalization.	Implement **Data Augmentation** (rotation, flipping) to expand training set.
3	Estimation of soil macronutrients (N, P, K) using deep learning and computer vision, 2022	ScienceDirect	ResNet-50 / Transfer Learning	Automates NPK detection without lab reagents.	Indirect correlation (colour vs. chemical) is weak.	Integrate **Domain Knowledge Injection** (soil type metadata) to improve correlation.
4	Soil Nutrient Detection Using Deep CNN and Multi-Spectral Image Processing, 2024	IEEE Access	CNN + multi-spectral input	Captures bands invisible to human eye (NIR).	Requires expensive multi-spectral cameras (not standard).	Focus on **RGB-to-Spectral Reconstruction** algorithms to use standard cameras.
5	Visible-near infrared spectroscopy and deep learning for soil organic carbon mapping, 2023	Elsevier	1D-CNN on Vis-NIR spectra	Extremely precise for Organic Carbon (SOC).	High hardware cost; point-based (not spatial).	Develop a **Smartphone-based Image System** to replace expensive spectrometers.
6	Identification of soil pH levels using colour-based	Springer	ANN + Colour Moments	Simple, computationally light.	Colour-pH correlation is soil-type dependent.	Train separate **Soil-Type Specific Heads** in the model.

	image processing and neural networks, 2022					
7	A Hybrid DL Model for Soil Phosphorus Prediction Agriculture, 2023	IEEE	CNN-LSTM Hybrid	Captures spatial and sequential dependencies.	Complex architecture; slow inference on edge devices.	Use **Model Quantization** to ensure it runs on mobile devices.
8	Deep learning architectures for soil fertility assessment: A review and meta-analysis, 2024	ScienceDirect	Meta-analysis of various DL	Provides broad understanding of state-of-the-art.	No specific novel algorithm proposed.	We implement the **best-performing architecture (e.g., EfficientNet)** identified here.
9	Estimating soil nitrogen content through deep learning-based analysis of soil surface images, 2022	Elsevier	Deep Belief Networks (DBN)	Good at unsupervised feature learning.	Older architecture; often outperformed by modern CNNs.	Use modern **state-of-the-art CNNs (ConvNeXt)** for better accuracy.
10	Smart Soil Health Monitoring System using IoT and Deep Learning, 2023	IEEE Xplore	IoT Sensors + CNN fusion	Real-time monitoring of temp/moisture + visual.	High deployment cost (sensors + connectivity).	**Zero-hardware approach**: Use visual indicators for moisture instead of sensors.
11	Application of 1D-CNN and 2D-CNN for soil chemical property estimation, 2023	ScienceDirect	1D (Spectra) & 2D (Image)	Versatile; handles different data modalities.	Requires dual data sources (spectra + image).	Simplify to **2D-only (Image)** input for usability.

12	A CNN-Transformer hybrid model for soil nutrient prediction, 2025	Elsevier	ViT (Vision Transformer)	Captures long-range dependencies in texture.	Data-hungry; struggles with small datasets.	Use **Transfer Learning** from ImageNet to handle smaller soil datasets.
13	Real-time Soil Fertility Evaluation using Mobile Camera and Deep Learning, 2023	IEEE	MobileNetV2 (Lightweight)	Runs directly on smartphones; offline capability.	Lower accuracy compared to deeper networks.	Use **Knowledge Distillation** to train a small model from a large one.
14	Rapid Prediction of Soil Texture and Nutrients via Image-Based DLAutoencoder, 2022	MDPI	Autoencoder + Random Forest	Reduces noise; good feature extraction.	Two-stage process is harder to optimize end-to-end.	Use an **End-to-End Regression CNN** for streamlined training.
15	Non-destructive evaluation of soil organic matter via deep learning on digital photos, 2023	ScienceDirect	VGG-16	Standard, well-understood architecture.	Very heavy model (weights); slow on mobile.	Replace VGG with **EfficientNet-B0** for better speed/accuracy trade-off.
16	Soil Moisture Classification Using Computer Vision and Deep Learning Techniques, 2025	IEEE Xplore	AlexNet / Custom CNN	Effective for classifying "Dry" vs "Wet".	Classification only (no percentage estimation).	Change output layer to **Regression (Linear Unit)** for specific moisture % values.
17	A deep learning approach for soil water content estimation	Elsevier	Regression CNN	Estimates water content values directly.	Prone to overfitting on specific soil colours.	**Colour Normalization** preprocessing to standardize soil appearance.

	using digital images, 2022					
18	Predicting soil moisture using ResNet-50 and image processing, 2023	ScienceDirect	ResNet-50	Deep architecture captures subtle texture changes.	"Black box" issue; farmers don't trust unexplained AI.	Add **Grad-CAM Heatmaps** to visualize moisture patterns.
19	Development of a low-cost camera-based soil moisture sensor using CNN, 2024	IEEE	Raspberry Pi + Simple CNN	Low-cost hardware implementation.	Limited processing power on Pi for complex models.	Offload processing to **Cloud API** or optimize for **Mobile Edge**.
20	Deep learning for soil moisture forecasting: A systematic review, 2023	Elsevier	LSTM / GRU (Time-series)	Predicts future moisture based on past data.	Requires historical data log (not instant assessment).	Focus on **Instantaneous Assessment** (current state) via single image.
21	Assessment of soil compaction using image processing and deep Learning, 2022	Springer	Morphological Ops + CNN	Detects surface crusting and compaction.	Surface-only; cannot see sub-surface compaction.	Correlate **surface cracks/texture** with known compaction metrics.
22	CNN-Based Soil Water Content Prediction Model for Agricultural Land, 2023	IEEE Xplore	CNN with Dropout	Robust against overfitting.	Ignores environmental context (weather).	Add **Auxiliary Inputs** (user enters current weather) to model.
23	Deep learning for real-time soil moisture monitoring in precision irrigation, 2024	ScienceDirect	YOLO (Object Detection)	Detects wet/dry *regions* within an image.	Focuses on localization, not quantification.	Combine **YOLO (location)** with **Regression Head (value)**.

24	Integrating UAV imagery and deep learning for soil moisture mapping, 2021	Elsevier	Semantic Segmentation (U-Net)	Maps moisture across large fields.	Requires Drone/UAV (high barrier to entry).	Adapt algorithms for **Handheld/Ground-level** photos.
25	Soil surface roughness estimation using computer vision and deep learning, 2023	IEEE	3D-CNN / Stereo Vision	Estimates physical roughness accurately.	Requires stereo cameras or multiple angles.	Use **Depth-from-Defocus** or Monocular Depth Estimation techniques.
26	Multi-modal deep learning for soil moisture and temperature prediction, 2024	ScienceDirect	Multi-modal Fusion	Combines thermal + RGB data.	Thermal cameras are expensive additions.	Train on RGB only to infer properties usually seen in thermal (e.g., wet soil is darker).
27	Soil bulk density estimation using image-based deep learning, 2022	Elsevier	Texture Analysis + ANN	Links visual texture to density.	Bulk density is a volumetric property (hard to see).	Use **Pore Space Analysis** on surface images as a proxy.
28	Classification of Soil Saturation Levels using Transfer Learning, 2023	IEEE	InceptionV3	Excellent feature extraction from pre-training.	Large model size (~90MB).	Use **Model Pruning** to reduce size for app download.
29	A comparative study of DL models for soil moisture estimation, 2024	ScienceDirect	Comparison (VGG vs ResNet)	Establishes benchmarks.	No new methodology proposed.	Select the **winner (often ResNet)** and optimize it further.

30	Using thermal and RGB images for soil water status assessment via DL, 2022	Elsevier	RGB-Thermal Fusion	High accuracy due to temperature data.	Dependency on thermal sensor alignment.	**Exclusive RGB approach** to maximize accessibility for farmers.
31	Soil Type Classification using Deep Learning: A Survey, 2023	IEEE Xplore	Review	Summarizes classification techniques.	--	Use the **taxonomy hierarchy** defined here for our class labels.
32	Automated soil classification using convolutional neural networks, 2021	ScienceDirect	Custom CNN	Automates taxonomy (Clay, Loam, Sand).	Limited to broad classes (USDA triangle).	Expand classes to include **local/vernacular soil names**.
33	Deep learning for soil texture classification: Impact of image resolution, 2022	Elsevier	CNN (varying inputs)	Analyses how pixel count affects accuracy.	Found high-res is needed (slow upload).	Use **Patch-based processing** (analyse crops, not full HD image).
34	Mobile-based Soil Classification System using CNN for Farmers, 2023	IEEE	TensorFlow Lite	Deployment-ready methodology.	UI/UX often neglected in research papers.	Focus on **User-Centric Design** (simple Red/Green/Yellow indicators).
35	An ensemble deep learning approach for soil classification, 2024	Springer	Ensemble (VGG + ResNet)	Higher accuracy than single models.	High computational cost (runs 2+ models).	Use **Snapshot Ensembles** (cheaper training) or a single robust model.
36	Identification of soil orders using deep learning and environment	ScienceDirect	DL + Geo-coordinates	Uses location to aid classification.	Fails if GPS is inaccurate or missing.	Make GPS **optional**, relying primarily on visual features.

	al covariates, 2022					
37	Transfer Learning for Soil Type Recognition in Diverse Climates, 2023	IEEE	MobileNet (Transfer)	Works well with limited local data.	Domain shift issues (Tropical vs Arid soil).	**Fine-tune** the model specifically on the target region's soil dataset.
38	Classification of soil horizons using deep learning and image analysis, 2023	Elsevier	Segmentation (Mask R-CNN)	Identifies layers in soil profile pits.	Requires digging a pit (destructive).	Focus on **Topsoil Assessment** (non-destructive).
39	Edge-computing for real-time soil classification using lightweight CNNs, 2024	ScienceDirect	SqueezeNet	Ultra-fast inference on edge.	Lower accuracy than larger models.	Balance with **MobileNetV3** (better accuracy/speed balance).
40	Comparative Analysis of VGG, ResNet, and Inception for Soil Classification, 2023	IEEE	Benchmarking	clear performance metrics.	--	Use these metrics to **justify our model choice** in the final report.
41	Deep learning in soil science: A comprehensive review, 2023	ScienceDirect	Review	Broad overview.	--	Use for **Literature Review** chapter drafting.
42	Federated Learning for Privacy-Preserving Soil Data	IEEE Xplore	Federated Averaging	Keeps data on user device (privacy).	Complex infrastructure setup.	Start with **Centralized Training**, move to on-device inference for privacy.

	Analysis, 2024					
43	Generative Adversarial Networks (GANs) for augmenting soil datasets, 2022	Elsevier	DCGAN / CycleGAN	Creates synthetic soil images.	Synthetic images may lack realistic texture.	Use **StyleGAN** for high-fidelity texture generation.
44	Explainable AI (XAI) for soil health assessment, 2024	ScienceDirect	LIME / SHAP	Explains *why* soil is "unhealthy".	Computationally expensive to generate.	Use **Class Activation Maps (CAM)** which are faster to compute.
45	Attention-based CNN for soil feature extraction, 2023	IEEE	CBAM (Attention Module)	Focuses on relevant soil parts (ignores background).	Adds complexity to the model.	Use **Spatial Attention** to ignore grass/rocks in soil images.
46	Deep reinforcement learning for optimal soil management, 2025	Elsevier	DRL (RL Agents)	Recommends actions (fertilizer amount).	Needs a simulated environment to train.	Focus on **Assessment first**, recommendation second (Rule-based).
47	Vision Transformers (ViT) for soil property mapping, 2024	ScienceDirect	ViT	State-of-the-art accuracy.	Needs massive data to beat CNNs.	Use **CNNs** initially as they are more data-efficient.
48	Robustness of Deep Learning models to soil image noise, 2023	IEEE	Denoising Autoencoders	Handles blurry/noisy images.	Pre-processing step adds latency.	Train model with **Gaussian Noise** added to become robust natively.
49	Cross-domain transfer learning for soil health prediction, 2022	Elsevier	Domain Adaptation	Applies lab data models to field photos.	"Lab-to-Field" gap is huge (lighting/texture).	Collect **paired datasets** (Lab result + Field Photo) for ground truth.

50	Hyperspectral image analysis for soil health via 3D-CNN, 2023	Springer	3D-CNN	Utilizes spectral depth.	Hyperspectral data is rare.	Stick to **RGB** (Red-Green-Blue) for universal compatibility.
51	Self-supervised learning for soil image representation, 2024	IEEE	SimCLR / MoCo	Learns from unlabeled data.	Training is computationally intensive.	Use **Pre-trained weights** from agricultural datasets if available.
52	Soil degradation detection using deep learning and satellite imagery, 2023	ScienceDirect	Change Detection CNN	Monitors soil over time (erosion).	Needs time-series images of same spot.	Enable **"Field History"** feature in app to track changes.
53	Deep learning for soil salinity mapping in arid regions, 2022	Elsevier	CNN (Salinity specific)	Detects white salt crusts effectively.	Confuses salt with sand or reflection.	Train on **hard negatives** (sand vs salt) to reduce false positives.
54	Graph Neural Networks for soil-plant interaction modelling, 2024	IEEE	GNN	Models complex relationships.	Requires graph-structured data.	Not applicable to pure **image-based classification**.
55	Uncertainty quantification in deep learning-based soil prediction, 2023	ScienceDirect	Bayesian CNNs	Gives confidence score (e.g., "80% sure").	Bayesian NNs are hard to train.	Use **Monte Carlo Dropout** at inference for uncertainty estimation.
56	UAV-based soil health monitoring using deep learning, 2021	Elsevier	Drone Imagery + CNN	Covers large areas quickly.	Low resolution per inch of soil.	**Close-up Smartphone Macro shots** for fine texture analysis.

57	Smartphone-based soil nutrient analysis: A deep learning approach, 2023	IEEE	Android/iOS + CNN	Democratizes soil testing.	Variability in phone camera sensors.	**Calibrate** using a standard reference card (e.g., color checker).
58	Deep learning for soil erosion susceptibility mapping, 2024	ScienceDirect	CNN + GIS Data	Large scale risk assessment.	Not for point-specific health checks.	Focus on **Point-diagnosis** for individual farmers.
59	Impact of data augmentation on soil property prediction models, 2022	Elsevier	Statistical Analysis	Proves augmentation helps.	--	**Aggressive Augmentation** is a core part of our pipeline.
60	Advances in AI-based soil health assessment: A review, 2025	IEEE Xplore	Systematic Review	Summary of future directions.	--	Use the **"Future Scope"** section to justify our project's relevance.

Table 7.2: Compiled Literature survey of the existing 60 research papers that we have considered.

Cluster / Research Theme	Papers Covered	Core Focus	Common Methodologies	Key Advantages Identified in Literature	Gaps / Limitations Identified	How Our Project Addresses This
Image-Based Soil Property Estimation (RGB Vision Models)	1, 2, 3, 6, 9, 14, 15, 17, 18, 22, 29, 33	Predict soil properties from digital images	CNN, ResNet, VGG, ANN, Regression CNN	Non-destructive, low cost, scalable, smartphone-friendly	Sensitive to lighting, color bias, dataset dependency	Robust preprocessing, augmentation, EfficientNet backbone
Spectral / Vis-NIR / Hyperspectral Techniques	4, 5, 11, 50	Chemical property estimation via spectral bands	1D-CNN, 3D-CNN, Multi-spectral fusion	Extremely precise chemical inference	Expensive hardware, low accessibility	RGB-only universal system design
Soil Moisture & Water Content Prediction	16, 17, 18, 19, 20, 22, 23, 24, 26, 28, 29, 30	Moisture estimation & classification	CNN, ResNet-50, YOLO, U-Net, LSTM/GRU	Strong visual correlation, good DL performance	Binary classification dominance, hardware fusion reliance	Regression-based indicators + explainability
Soil Classification & Texture Recognition	31, 32, 33, 34, 35, 36, 37, 38, 39, 40	Soil taxonomy & texture analysis	CNN, Transfer Learning, Ensemble, Patch-based CNN	High classification accuracy, strong feature learning	Broad classes only, large models	Region-aware classification + EfficientNet efficiency
Lightweight / Mobile / Edge AI Systems	13, 34, 39, 57	Smartphone / edge deployment	MobileNet, SqueezeNet, TFLite, Quantization	Offline inference, fast performance	Accuracy vs size trade-off	EfficientNetV2-B0 compromise architecture
Explainable AI (XAI) & Model Trust	18, 44	Interpretability & decision explanation	Grad-CAM, CAM, LIME, SHAP	Improves trust & transparency	Computational overhead (SHAP/LIME)	Fast Grad-CAM visual explanations
Data Augmentation, Robustness & Representation Learning	43, 48, 51, 59	Dataset limitations & noise handling	GANs, Denoising Autoencoders, SimCLR, Augmentation	Improves generalization	Synthetic realism issues, compute cost	Aggressive augmentation + noise-robust training
Multi-Modal / IoT / UAV / Sensor-Driven Systems	10, 24, 26, 30, 56	Fusion of sensors & imagery	IoT Fusion, UAV CNN, Thermal-RGB Fusion	Rich context, real-time monitoring	High deployment cost, hardware dependency	Zero-hardware visual inference approach
Soil Nutrient & Chemical Property Prediction	3, 4, 7, 9, 12, 14, 17, 53	Nutrient / SOM / chemical estimation	CNN-LSTM, ViT, DBN, Autoencoders	Automates lab-style inference	Weak visual-chemical correlations	Indicator-based reasoning instead of exact values
Advanced Architectures (Transformers / Attention / Hybrid DL)	7, 12, 45, 47	Improve feature representation	CNN-LSTM, ViT, Attention CNN (CBAM)	Captures long-range dependencies	Data-hungry, complex models	CNN-centric efficient architecture
Physical Soil Property Estimation (Density / Roughness / Compaction)	21, 25, 27, 55	Infer physical soil characteristics	Morphological Ops, 3D-CNN, ANN	Links texture to physical properties	Subsurface ambiguity, hardware needs	Surface-proxy visual indicators
Learning Paradigms & Training Strategies	42, 46, 49, 51	Federated / RL / Domain Adaptation / SSL	Federated Averaging, DRL, Domain Adaptation	Novel ML paradigms	Infrastructure complexity	Centralized training + robust inference
Large-Scale Monitoring & Remote Sensing	24, 52, 56, 58	Field-scale soil assessment	UAV CNN, Satellite CNN, Change Detection	Covers large regions	Not suited for point diagnosis	Focus on point-level farmer usability
Comparative / Benchmark / Review Studies	8, 29, 31, 40, 41, 60	Evaluate / summarize DL methods	Meta-analysis, Benchmarking	Establish performance baselines	No novel models	Justifies EfficientNetV2 selection

7.2 LITERATURE SURVEY (DETAILED)

Soil analysis using computer vision and deep learning has gained considerable attention due to its potential for non-destructive, low-cost, and scalable agricultural assessment. Numerous studies have investigated visual inference of soil properties, moisture estimation, nutrient prediction, classification, and model robustness. This section reviews key contributions in this domain.

Deep learning techniques have demonstrated promising performance in predicting soil properties directly from RGB images. In ***"Deep Learning-Based Prediction of Soil Organic Matter Using Digital Images"*** **(2023)**, the authors proposed a convolutional neural network (CNN) model trained on soil surface images to estimate soil organic matter (SOM). Their approach highlighted the feasibility of replacing chemical testing with image-based inference, achieving cost efficiency and non-destructive evaluation. However, the model exhibited sensitivity to illumination variations, emphasizing the need for lighting-invariant preprocessing techniques.

Similarly, the study ***"Convolutional Neural Networks for Predicting Soil Properties from RGB Images"*** (2021) employed multi-channel CNN architectures for predicting soil texture-related attributes. The authors demonstrated high accuracy for texture classification, particularly in sand and clay estimation. Nonetheless, the work stressed that large and diverse datasets are essential for robust generalization, indicating a common challenge in agricultural vision datasets.

Transfer learning has also been explored for soil chemical inference. In ***"Estimation of Soil Macronutrients (N, P, K) Using Deep Learning and Computer Vision"*** (2022), ResNet-50 was utilized to predict macronutrient levels. While the study demonstrated the capability of deep networks to capture visual patterns associated with nutrients, it acknowledged that correlations between soil colour and chemical composition remain indirect and unstable. This limitation motivates alternative formulations such as indicator-based reasoning rather than direct chemical regression.

Spectral imaging techniques provide an alternative pathway for soil property estimation. The work ***"Soil Nutrient Detection Using Deep CNN and Multi-Spectral Image Processing"*** (2024) combined CNNs with multi-spectral inputs, capturing bands beyond human-visible wavelengths. Although high predictive precision was achieved, the dependence on expensive multi-spectral cameras limits practical deployment. Comparable observations were made in ***"Visible–Near Infrared Spectroscopy and Deep Learning for Soil Organic Carbon Mapping"*** (2023), where Vis–NIR spectral data and 1D-CNN models yielded highly accurate predictions at the cost of specialized hardware requirements.

Soil moisture estimation constitutes a major research direction. In ***"A Deep Learning Approach for Soil Water Content Estimation Using Digital Images"*** (2022), regression CNN models were proposed for continuous moisture estimation. The study confirmed that visual cues contain sufficient information for predicting water content but noted risks of overfitting to soil colour distributions. Addressing similar objectives, ***"Predicting Soil Moisture Using ResNet-50 and Image Processing"*** (2023) leveraged deep residual networks for improved feature extraction, though interpretability concerns were raised due to the black-box nature of predictions.

To enhance spatial understanding, object detection and segmentation frameworks have been investigated. ***"Deep Learning for Real-Time Soil Moisture Monitoring in Precision Irrigation"*** (2024) adopted YOLO-based detection models to localize wet and dry regions within images. While localization accuracy improved, the approach focused primarily on classification rather than quantitative estimation, leaving room for hybrid regression–detection architectures.

Soil classification and taxonomy prediction have been extensively studied. In ***"Automated Soil Classification Using Convolutional Neural Networks"*** (2021), custom CNN architectures successfully categorized soils into major texture classes. Despite encouraging accuracy, the study was limited to broad USDA-based categories. Later works such as ***"Deep Learning for Soil Texture Classification: Impact of Image Resolution"*** (2022) revealed that classification accuracy strongly depends on image resolution, introducing computational and transmission challenges for high-resolution inputs.

Mobile and edge-compatible soil analysis systems have been explored to improve accessibility. The study ***"Real-Time Soil Fertility Evaluation Using Mobile Camera and Deep Learning"*** (2023) employed MobileNetV2 for lightweight inference directly on smartphones. Although efficient, lightweight architectures typically sacrifice accuracy compared to deeper networks. Similar trade-offs were observed in ***"Edge-Computing for Real-Time Soil Classification Using Lightweight CNNs"*** (2024), where SqueezeNet enabled fast inference with reduced predictive performance.

Explainable AI (XAI) methods have been proposed to address model transparency issues. In ***"Explainable AI for Soil Health Assessment"*** (2024), techniques such as LIME and SHAP were applied to interpret model decisions. While interpretability improved, computational overhead remained a significant limitation. Grad-CAM-based visualization techniques offer a computationally efficient alternative for highlighting discriminative regions contributing to predictions.

Data scarcity and environmental variability have motivated research into augmentation and robustness strategies. ***"Generative Adversarial Networks for Augmenting Soil Datasets"*** (2022) utilized GAN-based synthesis to expand training data. Despite improvements in dataset size, ensuring realistic texture generation remains challenging. Complementary research in ***"Robustness of Deep Learning Models to***

Soil Image Noise" (2023) emphasized denoising autoencoders and noise-aware training for enhancing model stability.

Advanced architectures including Vision Transformers (ViT) have recently been introduced. In ***"A CNN-Transformer Hybrid Model for Soil Nutrient Prediction"*** (2025), transformer-based attention mechanisms captured long-range dependencies in soil textures. However, transformer models generally require large datasets and high computational resources, making CNN-centric approaches more suitable for constrained agricultural scenarios.

Large-scale soil monitoring studies using UAV and satellite imagery have also been reported. ***"Integrating UAV Imagery and Deep Learning for Soil Moisture Mapping"*** (2021) demonstrated field-scale analysis but required aerial platforms and infrastructure. These approaches, while valuable for regional assessment, are less practical for point-level farmer diagnostics.

Uncertainty estimation has emerged as an important consideration. ***"Uncertainty Quantification in Deep Learning-Based Soil Prediction"*** (2023) introduced Bayesian CNN frameworks to produce confidence-aware predictions. Although theoretically sound, Bayesian networks often introduce training complexity, motivating lightweight alternatives such as Monte Carlo Dropout.

7.3 Key Observations from Literature

Across the surveyed works, several common challenges persist:

- Sensitivity to lighting and imaging conditions
- Hardware dependency in spectral approaches
- Weak visual–chemical correlations
- Accuracy–efficiency trade-offs in mobile models
- Limited interpretability and trust
- Dataset limitations and domain shifts

These limitations collectively justify the design of a robust, RGB-only, computationally efficient, and explainable soil health assessment framework.

8. PROPOSED METHODOLOGY

The proposed system presents a non-destructive soil health assessment framework using RGB soil images. The methodology integrates deep learning-based feature extraction with rule-based inference to estimate soil characteristics without requiring specialized sensors or laboratory analysis. The complete workflow is divided into the following stages.

8.1 Input Acquisition

The system accepts soil surface images captured using standard RGB cameras. Images may be obtained via mobile devices or datasets. No spectral or hardware-based sensors are required, ensuring low-cost deployment.

8.2 Image Preprocessing

Input images are standardized to ensure consistent model behaviour. Preprocessing includes resizing, normalization, and optional illumination correction techniques such as gamma adjustment and histogram equalization.

8.3 Soil Classification Using CNN

A convolutional neural network based on EfficientNetV2 is employed to classify soil types. The CNN learns discriminative texture and colour representations from images, producing a soil class prediction.

$$f_\theta(I) \rightarrow y_c$$

Where, I is the input image and y_c is the predicted class.

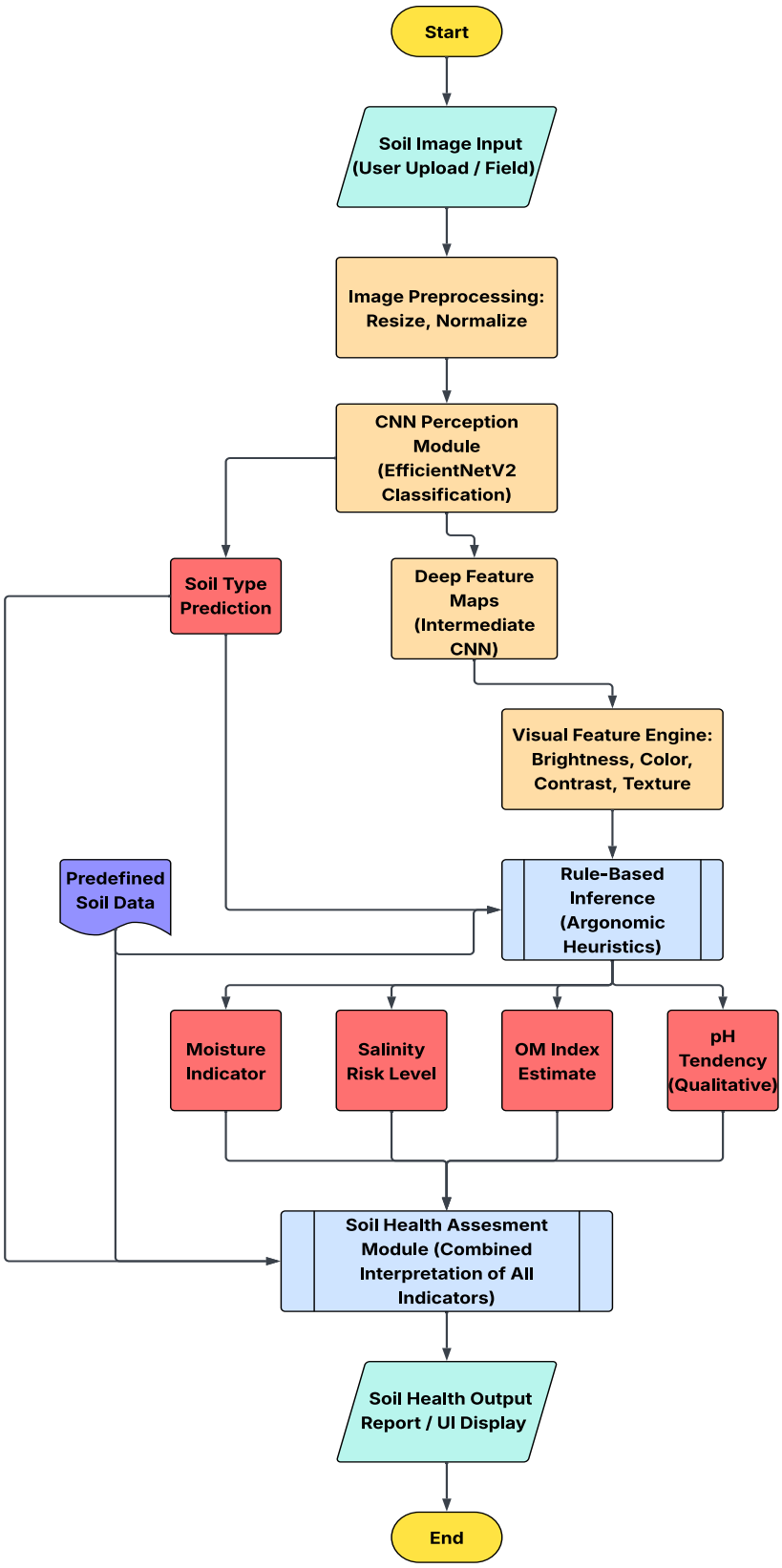


Fig: 7.1: Proposed Methodology (Block Diagram)

8.4 Deep Feature Extraction

Intermediate activations from the CNN's final convolutional layers are extracted. These features encode high-level visual descriptors necessary for downstream soil parameter estimation.

8.5 Rule-Based Parameter Estimation

Due to limited labelled datasets, soil attributes such as moisture, salinity, pH, and organic matter are inferred using domain-driven rules. The rule engine combines CNN features with image statistics to derive approximate estimates.

Example indicators include:

- Darker intensity → Higher moisture likelihood
- Whitish patches → Possible salinity presence
- Dark brown coloration → Organic matter cues

8.6 Soil Health Indicator Synthesis

Estimated attributes are aggregated into interpretable soil health indicators. The system prioritizes relative assessment rather than exact chemical prediction, improving robustness and usability.

8.7 Explainability Mechanism

Grad-CAM is integrated to generate visual explanation maps. These heatmaps highlight image regions contributing to predictions, improving transparency and user trust.

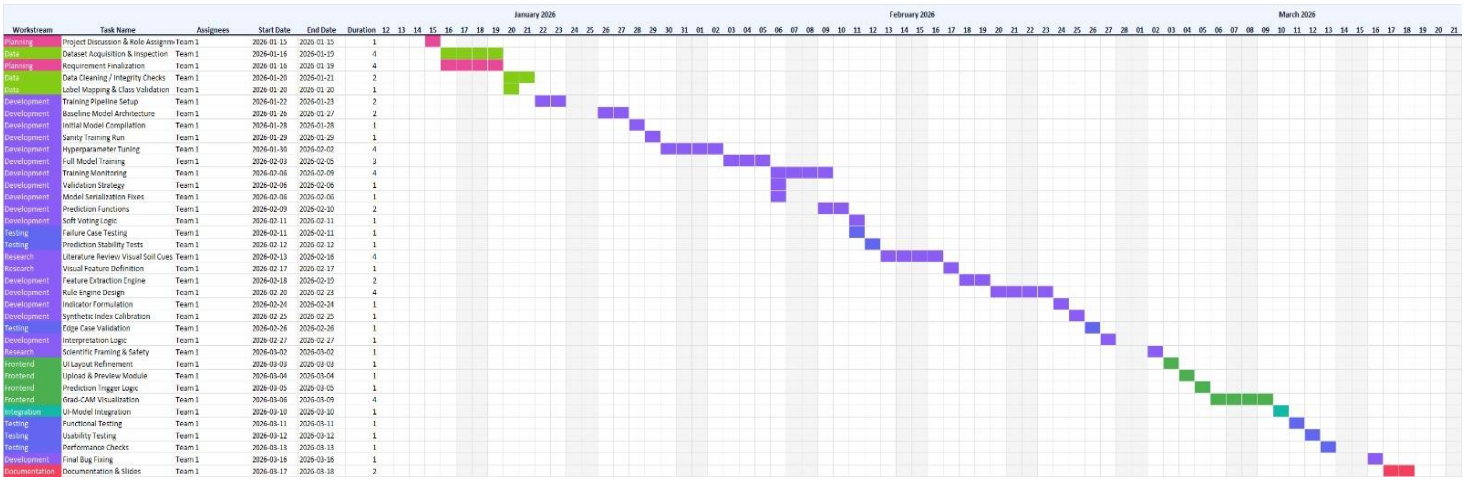
8.8 System Output

The system outputs:

- Soil classification result
- Estimated soil parameters
- Soil health indicators
- Confidence measures
- Visual explanation maps

9. PROJECT TIMELINE

9.1 Gantt Chart



Workstream	Task Name	Assignees	Start Date	End Date	Duration
Planning	Project Discussion & Role Assignm	Team 1	2026-01-15	2026-01-15	1
Data	Dataset Acquisition & Inspection	Team 1	2026-01-16	2026-01-19	4
Planning	Requirement Finalization	Team 1	2026-01-16	2026-01-19	4
Data	Data Cleaning / Integrity Checks	Team 1	2026-01-20	2026-01-21	2
Data	Label Mapping & Class Validation	Team 1	2026-01-20	2026-01-20	1
Development	Training Pipeline Setup	Team 1	2026-01-22	2026-01-23	2
Development	Baseline Model Architecture	Team 1	2026-01-26	2026-01-27	2
Development	Initial Model Compilation	Team 1	2026-01-28	2026-01-28	1
Development	Sanity Training Run	Team 1	2026-01-29	2026-01-29	1
Development	Hyperparamater Tuning	Team 1	2026-01-30	2026-02-02	4
Development	Full Model Training	Team 1	2026-02-03	2026-02-05	3
Development	Training Monitoring	Team 1	2026-02-06	2026-02-09	4
Development	Validation Strategy	Team 1	2026-02-06	2026-02-06	1
Development	Model Serialization Fixes	Team 1	2026-02-06	2026-02-06	1
Development	Prediction Functions	Team 1	2026-02-09	2026-02-10	2
Development	Soft Voting Logic	Team 1	2026-02-11	2026-02-11	1
Testing	Failure Case Testing	Team 1	2026-02-11	2026-02-11	1
Testing	Prediction Stability Tests	Team 1	2026-02-12	2026-02-12	1
Research	Literature Review Visual Soil Cues	Team 1	2026-02-13	2026-02-16	4
Research	Visual Feature Definition	Team 1	2026-02-17	2026-02-17	1
Development	Feature Extraction Engine	Team 1	2026-02-18	2026-02-19	2
Development	Rule Engine Design	Team 1	2026-02-20	2026-02-23	4
Development	Indicator Formulation	Team 1	2026-02-24	2026-02-24	1
Development	Synthetic Index Calibration	Team 1	2026-02-25	2026-02-25	1
Testing	Edge Case Validation	Team 1	2026-02-26	2026-02-26	1
Development	Interpretation Logic	Team 1	2026-02-27	2026-02-27	1
Research	Scientific Framing & Safety	Team 1	2026-03-02	2026-03-02	1
Frontend	UI Layout Refinement	Team 1	2026-03-03	2026-03-03	1
Frontend	Upload & Preview Module	Team 1	2026-03-04	2026-03-04	1
Frontend	Prediction Trigger Logic	Team 1	2026-03-05	2026-03-05	1
Frontend	Grad-CAM Visualization	Team 1	2026-03-06	2026-03-09	4
Integration	UI-Model Integration	Team 1	2026-03-10	2026-03-10	1
Testing	Functional Testing	Team 1	2026-03-11	2026-03-11	1
Testing	Usability Testing	Team 1	2026-03-12	2026-03-12	1
Testing	Performance Checks	Team 1	2026-03-13	2026-03-13	1
Development	Final Bug Fixing	Team 1	2026-03-16	2026-03-16	1
Documentation	Documentation & Slides	Team 1	2026-03-17	2026-03-18	2

Fig 8.1: Project Gantt Chart

Due to the size, it is not possible to accommodate the entire Gantt chart here in the document. The below link directs you to an excel file containing the Gantt Chart in detail.

Gantt Chart Link: https://docs.google.com/spreadsheets/d/12SYzY3xIC1Y8-8VXXv6wr_PDzD7tij2u/edit?usp=sharing&ouid=109120780670278079614&rtpof=true&sd=true

10. COMPARISON WITH EXISTING WORK

Several existing solutions use machine learning and deep learning for soil property estimation. Classical methods using Random Forest and SVR require manual feature engineering and are sensitive to lighting conditions. CNN-based models improve performance but are typically limited to single-parameter estimation and function as black-box systems.

Smartphone-based systems often require additional sensors or cloud-only processing, making them impractical for rural deployment. Moreover, very few systems provide a comprehensive soil health overview from a single input.

The proposed product differentiates itself by offering:

- End-to-end deep learning without handcrafted features
- Multi-parameter soil health inference
- Explainable predictions using visual heatmaps
- Robust performance under diverse field conditions
- Smartphone-first, offline-capable deployment

11. MOTIVATION AND FEASABILITY ANALYSIS

11.1 Technical Feasibility

Advancements in deep learning and mobile AI make camera-based soil analysis technically viable. Prior studies validate the correlation between visual soil patterns and physical-chemical properties.

11.2 Economic Feasibility

The system eliminates laboratory costs and additional hardware, making it affordable and scalable.

11.3 Social Feasibility

The product empowers farmers with instant soil insights, improving decision-making and digital inclusion.

11.4 Environmental Feasibility

Accurate soil assessment reduces overuse of fertilizers and supports sustainable agriculture.

11.5 Political and Policy Feasibility

The system aligns with national initiatives such as digital agriculture and soil health programs.

11.6 Demographic Feasibility

Designed for rural and semi-urban regions with high smartphone penetration and limited lab access.

12. PRODUCT-ORIENTED DESIGN CONSIDERATION

- Camera-only operation (no external sensors)
- Lightweight models for edge inference
- Offline functionality
- Simple visual indicators for usability
- Region-specific fine-tuning capability

13. PROJECT OUTCOME

12.1 Nature of Outcome

Product with Patent Potential

12.2 Expected Deliverables

- Functional soil health assessment application
- Optimized deep learning inference model

- Explainable and interpretable output system
- Documentation for patent filing

12.3 Application Areas

- Precision agriculture
- Fertilizer recommendation systems
- Soil monitoring for small and medium farms
- Government agricultural schemes

13. CONCLUSION

This project presents a novel, product-focused approach to soil health assessment using deep learning and camera-based image processing. By eliminating dependency on laboratories and specialized hardware, the proposed system offers a scalable, affordable, and explainable solution suitable for real-world agricultural deployment. The system not only addresses critical gaps in existing solutions but also establishes a strong foundation for commercialization and patenting.